

SECTION A, Question 1, COMPULSORY

- (a) Solve the initial value problem [2 marks]

$$\frac{dy}{dx} = 3x^2 e^{-y} y(0) = 1$$

- (b) With initial condition $x(0) = 0.5$, find the solution of the differential equation [3 marks]

$$t \frac{dx}{dt} + x = t$$

- (c) A body at an unknown temperature is placed in a room which is held at a constant temperature of $30^\circ F$. If after 10 minutes the temperature of the body is $0^\circ F$ and after 20 minutes the temperature of the body is $15^\circ F$, find the unknown initial temperature. [3 marks]

- (d) Find the inverse Laplace transform of [2 marks]

$$\frac{1}{s^2 + s}$$

- (e) Evaluate the line integral of the vector field $\vec{F} = 5xz\vec{i} + (3x^2 + 2y)\vec{j} + x^2z\vec{k}$ along the path from $(0, 0, 0)$ to $(1, 1, 1)$ parameterized by (t, t^2, t) . [3 marks]

- (f) Find the magnitude of the gradient for the scalar field $f(x, y, z) = x^2 + 3y^2 + z^3$ at the point $(1, 1, 1)$. [3 marks]

- (g) Find the value of $y(1)$ given that $y(x)$ is the solution of the differential equation with initial conditions $y(0) = 0$ and $y'(0) = 1$. [4 marks]

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 4y = 0$$

$$t \frac{dx}{dt} = t \Rightarrow dx = dt$$

$$\frac{1}{2} = \frac{C}{0} \Rightarrow C = 0$$

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

SECTION B, Attempt and TWO Questions

Q2 (a) Use Laplace transform to solve the initial value problem [7 marks]

$$\frac{d^2 x}{dt^2} + x = 2t, \quad x(0) = 0, \quad x'(0) = 5$$

(b) Use Green's formula to evaluate the following line integral around the sides of the triangle with vertices $A(0,0), B(1,0), C(1,2)$. [8 marks]

$$\oint x y dx + x^2 y^3 dy$$

Q3 (a) Represent the region of integration, change the order of integration, and evaluate the integral [9 marks]

$$\int_0^1 \int_{x^2}^x (1+2y) dy dx$$

(b) Verify whether the following vector field is irrotational [6 marks]

$$\vec{V} = x^2 \vec{i} + \frac{1}{2} y^2 \vec{j} + \frac{1}{2} z^2 \vec{k}$$

Q4 (a) Find the Laplace transform of $e^{-t} f(t)$ when the Laplace transform of $f(t)$ is given as $F(s) = \frac{3s^2-1}{s^2+s+1}$. [5 marks]

(b) Obtain the general solution of the differential equation by using the method of undetermined coefficients and deduce the particular solution satisfying $y(0) = 1$ and $y'(0) = 0$. [10 marks]

$$\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 3y = 10 \cos 2x$$

.....END.....

$$x \pm ip$$

2

$$2 \cdot y(0) - 3 y'(0) -$$

$$s^2 x(s) - s x(0) - x'(0)$$

$$(-i)^2$$